PhD Position: Engineering of Self-Transcendent Experiences in VR

Laboratory of Cognitive Neuroscience (LNCO), EPFL, Campus Biotech, Geneva
PIs: Prof. Olaf Blanke, Dr Bruno Herbelin | Duration: 4 years | Start: Feb 2027

The project

Self-transcendent experiences, marked by awe, a sense of connectedness and a loosening of the boundaries of the self, can durably improve well-being. They are known mainly from psychedelic research, but are hard to study in a controlled way. This project develops a non-pharmacological "cyberdelic" alternative that combines immersive virtual reality with simulated weightlessness.

You will lead three experimental studies with healthy volunteers. **Study 1:** a within-subject study of awe and out-of-body perspective shifts in a cinematic VR scenario, the "Virtual Overview" inspired by astronautes' experiences. **Study 2:** a comparison methods for the induction of weightlessness (a robotic vestibular motion platform, a dry flotation bed and aquatic VR in a swimming pool), using a new psychophysical task measuring perceived hand weight. **Study 3:** an integrated study combining the VR scenario and weightlessness induction (aquatic VR), assessed with psychometric scales, behavioural measures, physiology and interviews.

You will also contribute to a co-design study with patients and clinicians at Geneva University Hospitals (HUG) on the potential of this approach in palliative care. The project is carried out in partnership with HUG, NiGHT IMMeRSION S.A., Birkbeck University of London, and John Moores University.

What we are looking for

- **Advanced VR development skills** (Unity3D or similar), able to build rich, complex scenarios combining 3D environments, avatars, camera effects and spatial audio.
- **Technical aptitude for hardware control**, including driving and synchronising a robotic vestibular platform with VR and physiological recording.
- **A strong interest in experimentation with human participants**: designing protocols, recruiting, testing and following ethical and safety procedures.
- **An interest in data analysis**, including statistical modelling (e.g. mixed-effects models), physiological signals and the development of **AI-based semantic analysis** of interview transcripts.
- **Collaboration skills**: you will work with cognitive neuroscientists, engineers, artists, clinicians and a company across several sites and disciplines.
- **Human qualities**: empathy, tact and respect for patients and carers, as the project is aimed at palliative care.

A Master's degree in neuroscience, computer science, engineering, psychology or a related field is required. Experience in VR, bodily self-consciousness, psychophysics or contemplative practices is a plus. A commitment to open science (open code, pre-prints, open access) is expected.

How to apply

The successful candidate must be admitted to the **EPFL doctoral program in Robotics, Control and Intelligent Systems (EDRS)** and must apply by the **15 December 2026 deadline**: https://www.epfl.ch/education/phd/edrs-robotics-control-and-intelligent-systems/. When applying, please indicate Prof Blanke (LNCO) as your wish for host laboratory.

After successful admission and for more information on the project, you can contact bruno.herbelin@epfl.ch.